One dimension array

Ex) Write a program to print one dimension array vertically (each element in a separate line)

```
#include <iostream.h>
Void main()
{ int a[10],i;
 for(i=0; i<10; i++)
  cin>>a[i];
for(i=0; i<10; i++)
  cout<<a[i]<<'\n';
}
```

Ex) Write a program to find the average of one dimension array

```
#include <iostream.h>
Void main()
{ int  a[10] , i , s=0 ;
 float  av=0.0;
 for(i=0; i<10; i++)
  cin>>a[i];
for(i=0; i<10; i++)
  s+=a[i];
av=s/10;
cout<<“average=”<<av;
}
```

Ex1) Write a program to find the number and multiplication of the elements which larger than the last element in one dimension array

Ex2) Write a program to inverse one dimension array

Arrays

Bubble sort

How to sort one dimension array ?

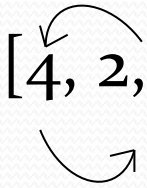
One of the most important algorthems to sort one dimension arrays is (bubble sort) which can be illustrated as in bellow :

Suppose that we have the following element:

[4, 2, 3, 6, 1]

And we want to sort it in ascending order to be

[1, 2, 3, 4, 6]



[4, 2, 3, 6, 1]

[2, 4, 3, 6, 1]
[2, 3, 4, 6, 1]
[2, 3, 4, 6, 1]
[2, 3, 4, 1, 6]

[2, 3, 4, 1]
[2, 3, 4, 1]
[2, 3, 1, 4]

[2, 3, 1]
[2, 1, 3]

[1, 2]

[1, 2, 3, 4, 6]

Bubble sort

```
#include<iostream.h>
#include<conio.h>
void main()
{
clrscr();
int a[6],i,j,x,n;
Cout<<“how many element:”;
Cin>>n;
Cout<<“now give your elements: “;
for(i=0;i<n;i++)
Cin>>a[i];
for(i=0;i<n-1;i++)
for(j=0;j<n-i-1;j++)
if(a[j]>a[j+1])
{x=a[j]; a[j]=a[j+1];a[j+1]=x;
}
for(i=0;i<n;i++)
Cout<<a[i]<<‘\n’;
getch();
}
```

Two Dimensional Arrays

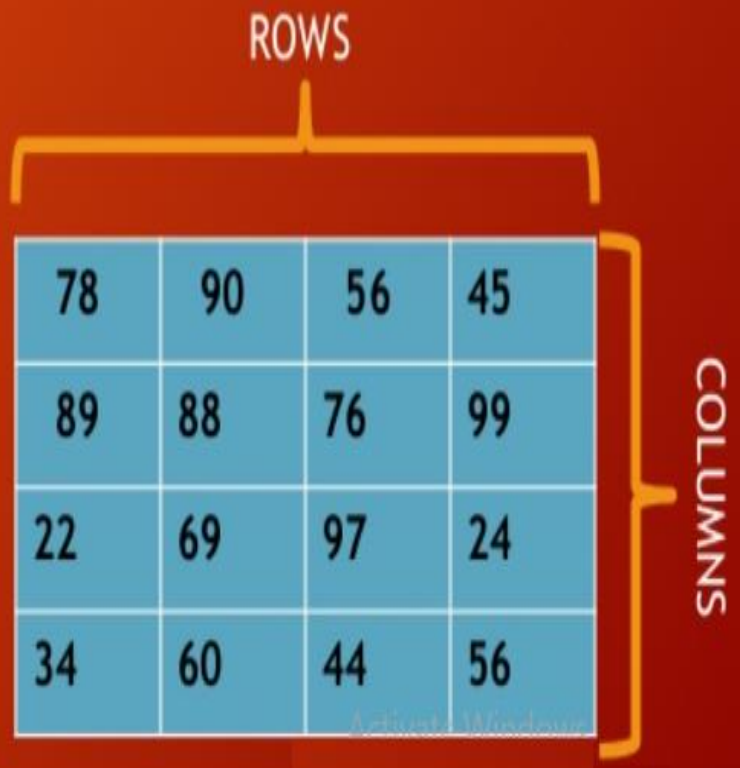
Contents

- Two-Dimensional Array
- Declaring Two-Dimensional Array
- Initializing Two-Dimensional Array
- Accessing elements of Two-Dimensional Array
- Inputting elements of Two-Dimensional Array
- Outputting elements of Two-Dimensional Array
- Operations on two Dimensional arrays

What is a Two Dimensional Array ?

- A two dimensional array is the arrangement of elements in rows and columns.
- It has two indices, one for row and other for column. First index represents the number of rows, second index represents the number of columns.

Example : int list[4][4];



Declaring a Two Dimensional Array

data-type name [no of rows] [no of columns];

integer constant/ literal

Example :

int List[4][4];

- Data type= integer
- Name= list
- No. of rows=4
- No. of columns=4

	0	1	2	3
0	List[0][0]	List[0][1]	List[0][2]	List[0][3]
1	List[1][0]	List[1][1]	List[1][2]	List[1][3]
2	List[2][0]	List[2][1]	List[2][2]	List[2][3]
3	List[3][0]	List[3][1]	List[3][2]	List[3][3]

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Initializing a Two Dimensional Array

- A two dimensional array can be initialized using curly braces:

row 1 row 2 row 3 row 4

```
int  List[4][2]={{20, 25},{22, 24}, {18, 22}, {17, 24}};
```

We can also initialize in this way:

```
int List[4][2]={20, 25, 22, 24, 18, 22, 17, 24};
```



Memory Representation

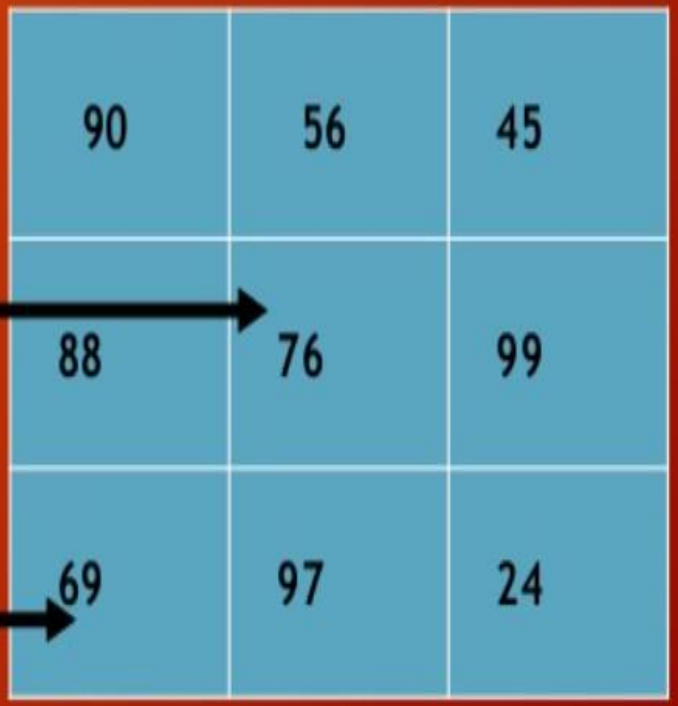
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Accessing two dimensional array elements

```
int a[3][3]={90,56,88,76,99,69,97.24};

cout<<[1][1];     // displays 76

cout<<[2][0];     // displays 69
```



Inputting values into a two Dimensional Array

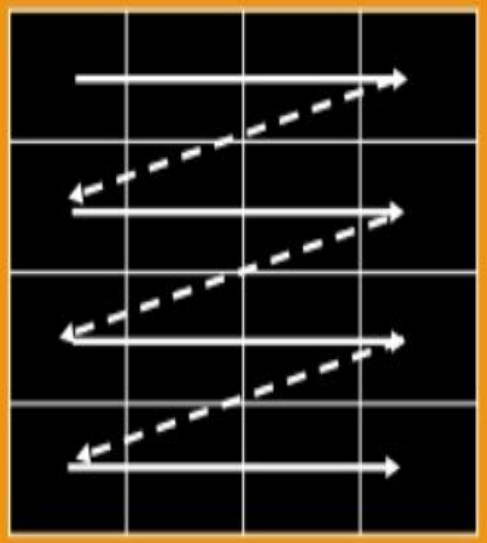
To input values into a two dimensional array we use a double loop.

Example : To input elements into the array List the form would be:

```
for(int i=0; i<4; i++)        outer loop for row index
  for(int j=0;j<2; j++)       inner loop for column index
     cin>>list[i][j];
```

The input order will be row wise.

Row wise input



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Ex) Write a program to read and print two dimension array

Traversal means visiting each element of the array once. An example program would be to input and display the elements.

```
# include<iostream.h>
void main()
{ int A[20][20]; // array declaration
int M, N; // for no of rows and columns to be input
cout<<"Enter no of rows and columns";
cin>>M>>N;
for (int i = 0; i < M; i++)
for (int j = 0; j < N; j++)
{
cout<<"\nEnter element at row & column "<<i<<" ,
"<<j<<":";
cin >> A[i][j];
 }
```

```
cout<<"\n\nThe elements of the array are :";
cout<<"\n **********\n";
for (int i = 0; i < M; i++)
{ for (int j = 0; j < N; j++)
 cout<< A[i][j]<<"\t";
 cout<<'\n';
  }
}
```

Ex) Write a program to find the summation of each row in two dimension array

2. Sum of row elements : Program

```
# include<iostream.h>
void main()
{ int A[20][20], M,N, sum;
int M, N;
cout<<“Ënter no of rows and columns”;
cin>>M>>N;
for (int i = 0; i < M; i++)
for (int j = 0; j < N; j++)
{
cout<<”\nEnter element at row & column ”<<i<<” , ”<<j<<”:”;
cin >> A[i][j];
 }
cout<<“\n”;
for(i=0;i<M;i++)
{
sum=0;
for(int j=0;j<N;j++)
sum+=A[i][j];
cout <<“\nSum of Row  "<< i +1 << “ ="<<sum;
}
```

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Two dimension array examples (1)

a[0,0]=2	a[0,1]=5	a[0,2]=7
a[1,0]=3	a[1,1]=0	a[1,2]=1
a[2,0]=7	a[2,1]=8	a[2,2]=2

Suppose the following

n=number of rows or columns

i=row sequence, j=column sequence

Then we can conclude the following

Location	condition
Primary diagonal	i=j
Upper Primary diagonal	i<j
Secondary diagonal	i+j=n-1
Under Secondary diagonal	i+j>n-1

The solution

Ex1) Write a program to read two dimension array then print it as rows and columns and find its summation

```
#include<iostream.h>
void main()
{int a[3][3],i,j,s=0;
for(i=0;i<3;i++)
for(j=0;j<3;j++)
cin>>a[i][j];

for(i=0;i<3;i++)
for(j=0;j<3;j++)
s+=a[i][j];

for(i=0;i<3;i++)
{
for(j=0;j<3;j++)
  cout<<a[i][j]<<"   ";
  cout<<'\n';
}
cout<<'\n'<<s;
}
```

The solution

Ex2) Write a program to find the summation of the primary diagonal in two dimension array

```
#include<iostream.h>
#include<conio.h>
void main()
{
clrscr();
int a[3][3],i,j,s=0;

for(i=0;i<3;i++)
for(j=0;j<3;j++)
cin>>a[i][j];

for(i=0;i<3;i++)
 s+=a[i][i];

cout<<"summation= "<<s;
getch();
}
```

Ex4) Write a program to find the multiplication of the following

1-first row
2- second column
3- second half of the array rows

The solution

```
#include<iostream.h>
#include<conio.h>
void main()
{
clrscr();
int a[4][4],i,j,m1=1,m2=1,m3=1;

for(i=0;i<4;i++)
for(j=0;j<4;j++)
cin>>a[i][j];

for(i=0;i<4;i++)
for(j=0;j<4;j++)
{
if (i==0)m1*=a[i][j];
if (j==1)m2*=a[i][j];
if (i>1)m3*=a[i][j];

 }
```

```
for(i=0;i<4;i++)
{
for(j=0;j<4;j++)
  cout<<a[i][j]<<"  ";

  cout<<'\n';
}
cout<<"first row="<<m1<<'\n';
cout<<"second column="<<m2<<'\n';
cout<<"second half="<<m3<<'\n';

getch();

}
```

Ex3) Write a program to find the summation of the following

1-primary diagonal
2- secondary diagonal
3- the elements upper the primary diagonal
4-the elements under the secondary diagonal

The solution

```
#include<iostream.h>
#include<conio.h>

void main()
{
clrscr();
int a[3][3],i,j,s1=0,s2=0,s3=0,s4=0;

for(i=0;i<3;i++)
for(j=0;j<3;j++)
cin>>a[i][j];

for(i=0;i<3;i++)
for(j=0;j<3;j++)
{
if (i==j)s1+=a[i][j];
if (i+j==2)s2+=a[i][j];
if (i<j)s3+=a[i][j];
if (i+j>2)s4+=a[i][j];
 }
```

```
for(i=0;i<3;i++)
{
for(j=0;j<3;j++)
  cout<<a[i][j]<<"  ";

  cout<<'\n';
}
cout<<"primary diagonal="<<s1<<'\n';
cout<<"secondary diagonal="<<s2<<'\n';
cout<<"upper primary diagonal="<<s3<<'\n';
cout<<"under secondary diagonal="<<s4;
getch();
}
```

Two dimension Array examples (2)

Ex1) Write a program to create the following two dimension array

1	1	1
2	2	2
3	3	3

```
# include <iostream.h>
Void main()
{  int a[3][3] ,i,j;

  for (i=0;i<3;i++)
            for (j=0;i<3;i++)
             a[i][j]=i+1;
for (i=0;i<3;i++)
{
            for (j=0;j<3;j++)
             cout<<a[i][j]<<“    ”;
     Cout<<‘\n’;
}
}
```

Ex2) Write a program to create the following two dimension array

1	2	2
3	1	2
3	3	1

```
# include <iostream.h>
Void main()
{  int a[3][3] ,i,j;

  for (i=0;i<3;i++)
            for (j=0;i<3;i++)
             if (i==j) a[i][j]=1; else if (i<j) a[i][j]=2;else a[i][j]=3;

for (i=0;i<3;i++)
{
            for (j=0;j<3;j++)
             cout<<a[i][j]<<“     ”;
     Cout<<‘\n’;
}
}
```

Ex3) Write a program to create the following two dimension array

1	2	3
4	5	6
7	8	9

```
# include <iostream.h>
Void main()
{  int a[3][3] ,i,j,k=0;

  for (i=0;i<3;i++)
            for (j=0;i<3;i++)
             {k++;
             a[i][j]=k;
            }
for (i=0;i<3;i++)
{
            for (j=0;j<3;j++)
             cout<<a[i][j]<<“    ”;
    Cout<<‘\n’;
}
}
```

Ex4) Write a program to create the following two dimension array

1	4	6
8	1	12
14	16	1

```
# include <iostream.h>
Void main()
{  int a[3][3] ,i,j,k=0;

  for (i=0;i<3;i++)
            for (j=0;i<3;i++)
             {k+=2;
             if (i==j) a[i][j]=1; else a[i][j]=k;
            }
for (i=0;i<3;i++)
{
            for (j=0;j<3;j++)
             cout<<a[i][j]<<“    ”;
    Cout<<‘\n’;
}
}
```